Estimating Energy Consumption

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1 Introduction

The energy efficiency of a digital signal processing (DSP) technique used in a coherent receiver cannot be judged from the receiver alone: a simpler implementation that saves receiver energy typically tolerates a lower SNR, which the transmitter must compensate for with extra launch power. A like-for-like comparison between architectures must therefore account for both contributions.

This section develops such a framework. The receiver contribution is modelled from operation counts of real additions and multiplications, with per-operation energies that depend on the fixed-point word length, allowing implementations to be compared without a hardware build. The transmitter contribution is derived from the minimum SNR each implementation requires to meet the pre-FEC BER threshold defined in the coherent PON (CPON) standard [1] under investigation. The two are combined for a downstream PON to give the per-bit system energy

2 Receiver Energy

The energy consumption of the receiver can be estimated from the number of real multiplications N_M and real additions N_A it performs to process each symbol. The true cost of these operations depends on the hardware implementation. Here the reference values shown in table 1 are used to allow for comparison between different DSP algorithms. An accurate estimate of their actual energy consumption would require analysis of their ASIC implementation for a modern process node.

Table 1: Energy cost of arithmetic operations for different bit widths in 45 nm CMOS [2].

n (bits)	Add (pJ)	Multiply (pJ)
8	0.03	0.2
32	0.1	3.1

2.1 Dependence on Word Length

A straightforward adder implementation performs n bit-wise additions on words of length n, so its energy scales linearly with n. Multiplication accumulates n partial products of length n and so scales as n^2. These physical scaling laws are imposed *a priori* and the proportionality constants calibrated by least-squares fit through the origin to the two data points of table 1, giving

$$E_A(n) = 3.16\, n \text{ fJ}, \qquad E_M(n) = 3.03\, n^2 \text{ fJ}. \tag{1}$$

2.2 Total Energy and Power

The energy per symbol for the entire receiver is given by

$$E_{\text{rx},s} = \sum_{k=1}^{L} 2\left[N_{A,k}E_A(n_k) + N_{M,k}E_M(n_k)\right], \tag{2}$$

where the receiver is decomposed into L blocks that use different word lengths n_k according to their required precision, and $N_{A,k}$, $N_{M,k}$ are the addition and multiplication counts per symbol. The factor of 2 is for both polarizations. Blocks operating on the oversampled signal at rate $f > R_s$ have their per-sample operation counts scaled by the oversampling ratio f/R_s. The receiver power is then

$$P_{\text{rx}} = R_s E_{\text{rx},s}, \tag{3}$$

and converting to energy per bit allows fair comparison across modulation schemes and symbol rates,

$$E_{\text{rx}} = \frac{E_{\text{rx},s}}{\log_2(M)}, \tag{4}$$

where M is the modulation order.

3 Transmitter Energy

The downstream link considered here consists of a single OLT transmitting through a 1:K passive splitter to K ONUs, as shown in fig. 1. The OLT power is divided equally between branches, so each ONU receives a $1/K$ share. This is a simplification of a typical PON where there may be several stages of passive splitters placed at different locations along the fibre, in which case the noise characteristics of the fibre between splitter stages must be analysed independently.

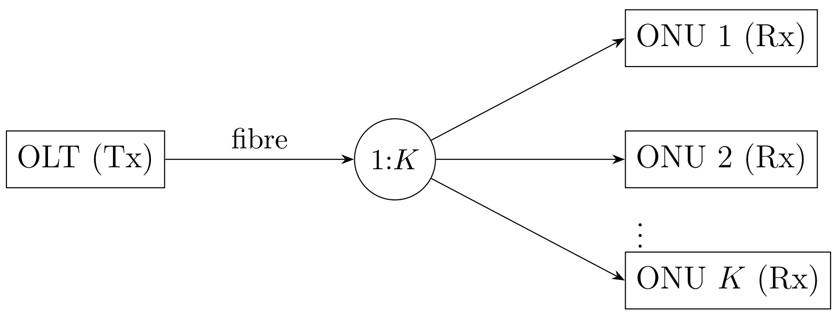

Figure 1: Downstream PON link: one OLT transmitting to K ONUs.

3.1 Minimum Required SNR

The CPON standard specifies a maximum pre-FEC bit-error rate (BER) of 10^{-2} that must not be exceeded for the forward error-correction (FEC) scheme to function correctly. Each receiver implementation therefore has a minimum required SNR to meet this FEC threshold. This SNR, together with the noise characteristics of the optical network, determines the minimum transmitter power.

3.2 Noise

3.2.1 Shot Noise

The relationship between SNR and transmitter power P_{tx} depends on the dominant source of noise in the network. Optical access networks typically operate over short lengths at low power, so amplifiers can be omitted and non-linear noise neglected, leaving shot noise as the dominant impairment. The shot noise power at the receiver is $2h\nu B$. After fibre attenuation and the 1:K splitter, the signal power reaching each ONU is $P_{\text{tx}} \cdot 10^{-\alpha L/10}/K$, giving a per-ONU SNR of

$$\text{SNR} = \frac{P_{\text{tx}} \cdot 10^{-\alpha L/10}}{2Kh\nu B}, \tag{5}$$

where B is the signal bandwidth, α is the fibre loss in dB/km, and L is the fibre length. For receiver implementations with different FEC SNRs, the corresponding difference in transmitter power is

$$\Delta P_{\text{tx}} = P_{\text{tx},1} - P_{\text{tx},2} = 2Kh\nu B \cdot 10^{\alpha L/10}(\text{SNR}_1 - \text{SNR}_2), \tag{6}$$

and the corresponding difference in optical transmit energy per bit is

$$\Delta E_{\text{tx}} = \frac{1}{R_s}\left(\frac{P_{\text{tx},1}}{\log_2(M_1)} - \frac{P_{\text{tx},2}}{\log_2(M_2)}\right), \tag{7}$$

where the symbol rate R_s is taken equal to the signal bandwidth B under Nyquist signalling. The general form admits different modulation orders M_1, M_2 for cross-format comparisons; for fixed M it reduces to $\Delta E_{\text{tx}} = \Delta P_{\text{tx}}/(R_s \log_2 M)$.

3.2.2 Amplifier and Non-Linear Noise

Long-reach access networks require EDFAs in the trunk fibre, in which case amplifier and non-linear noise can no longer be neglected. Each EDFA with gain G and noise figure NF (dB) contributes an ASE noise power

$$P_{\text{ASE}} = N_{\text{ASE}}B = 10^{\text{NF}/10}h\nu(G-1)B, \tag{8}$$

and assuming each amplifier exactly compensates the loss of its preceding span, every span re-launches with the same input power P_{tx}. The Gaussian Noise (GN) model [3] then gives the per-span non-linear interference (NLI) power as

$$P_{\text{NLI}} = \frac{C_{\text{NLI}}P_{\text{tx}}^3}{B^2}, \tag{9}$$

where C_{NLI} lumps the fibre dispersion, attenuation, span length, and non-linearity coefficient.

For a link of n identical spans, the ASE and NLI contributions from each span accumulate, giving a total noise power $n(P_{\text{ASE}} + P_{\text{NLI}})$. With the 1:$K$ passive splitter placed at the end of the trunk, both signal and accumulated noise are attenuated equally between the OLT and each ONU, so the splitter ratio cancels in the per-ONU SNR:

$$\text{SNR} = \frac{P_{\text{tx}}}{n\left(N_{\text{ASE}}B + \frac{C_{\text{NLI}}P_{\text{tx}}^3}{B^2}\right)}, \tag{10}$$

where P_{tx} is both the OLT transmit power and (under the equal-compensation assumption) the launch power into each span. The required transmit powers for each implementation are solved for numerically from eq. (10), with the difference in transmitter energy following from eq. (7).

3.3 Laser Efficiency

The energy difference ΔE_{tx} of eq. (7) is the difference in the laser's optical output energy. The corresponding difference in wall-plug energy consumed by the transmitter must account for the wall-plug efficiency η of the laser:

$$\Delta E'_{\text{tx}} = \frac{1}{\eta}\Delta E_{\text{tx}}. \tag{11}$$

4 System Energy

The total energy consumed to deliver one bit to every ONU is the sum of the wall-plug transmitter energy and the per-ONU receiver energy summed over all K ONUs of fig. 1,

$$E_{\text{sys}} = E'_{\text{tx}} + KE_{\text{rx}}, \tag{12}$$

where $E'_{\text{tx}} = E_{\text{tx}}/\eta$ is the wall-plug optical transmit energy per bit (eq. (11)) and E_{rx} is the per-ONU receiver energy per bit (eq. (4)). When comparing two receiver implementations, the change in system energy per bit is

$$\Delta E_{\text{sys}} = \Delta E'_{\text{tx}} + K\,\Delta E_{\text{rx}} = \frac{1}{\eta}\Delta E_{\text{tx}} + K\,\Delta E_{\text{rx}}, \tag{13}$$

where ΔE_{tx} follows from eq. (7) (with the explicit form depending on the dominant noise source) and ΔE_{rx} is the change in receiver energy per ONU from eq. (4).

References

[1] Cable Television Laboratories, Inc. *Coherent PON Physical Media Dependent Layer 1.0 Specification*. Tech. rep. CPON-SP-PMDv1.0-I01-251216. Issued specification, version I01. CableLabs, Dec. 2025. URL: `https://www.cablelabs.com`.

[2] Mark Horowitz. "Computing's Energy Problem (and what we can do about it)". In: *2014 IEEE International Solid-State Circuits Conference (ISSCC) Digest of Technical Papers*. 2014, pp. 10–14. DOI: `10.1109/ISSCC.2014.6757323`.

[3] Pierluigi Poggiolini et al. "Analytical Modeling of Nonlinear Propagation in Uncompensated Optical Transmission Links". In: *IEEE Photonics Technology Letters* 23.11 (2011), pp. 742–744. DOI: `10.1109/LPT.2011.2131125`.